LANTERNFIELD LABS

Project Keystone

REPORT

Title	**Sample size estimate (Task #4)**
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Date	September 18, 2026

Summary

Recruiting **60 participants** gives the study enough data to judge whether the sensor agrees with the lab reference method, with room for about 10 percent dropout.

Assumptions

Input	Value	Source
Expected mean difference (sensor minus lab)	0.5 units	Pilot data
Expected standard deviation of differences	2.0 units	Pilot data
Maximum acceptable difference	5.0 units	Clinical team
Power	90%	Study standard
Significance level	0.05	Study standard

Result

Step	Participants
Required for agreement analysis	54
Added for expected dropout (10%)	6
Total to enroll	**60**

Next steps

Use 60 as the enrollment target in the protocol and budget. Revisit after the dry run (Task #31) if the measured spread is larger than expected.

Fictional sample document created to demonstrate the Project Keystone tracker. Not a real study.